Lecture 15

Strong Continuity and Infinitesimal Generators

Objectives. *Part II studies unitary representations on infinite-dimensional Hilbert spaces. Their derivatives may exist only on a proper subspace. This lecture fixes the continuity hypothesis, defines the derivative on smooth vectors, and uses Stone's theorem to identify the skew-adjoint generator of each one-parameter subgroup, together with its domain. Translations on $L^2(\mathbb{R})$ are the model throughout.*

By the end of this lecture you will be able to:

- Prove strong continuity of a homomorphism into the unitary group from continuity at the identity on a dense set.
- Define smooth vectors and $d\Pi$, and prove that each $d\Pi(X)$ is skew-symmetric on smooth vectors.
- State Stone's theorem with its hypotheses, and identify the closure of $d\Pi(X)$ with the generator.
- Locate explicit vectors relative to the domains for translations: Schwartz functions, a non-Schwartz vector in the domain, and a vector outside it. Prove that translations have no nonzero finite-dimensional invariant subspace.

15.1 From matrices to Hilbert spaces

Let $\Pi\colon G \to \mathrm{GL}(V)$ be a continuous representation on a finite-dimensional complex space V. Continuous homomorphisms of Lie groups are smooth (Lecture 3). So every $X \in \mathfrak{g}$ gives an operator

$$d\Pi(X) = \frac{d}{dt}\Big|_{t=0} \Pi(\exp tX) \in \mathrm{End}(V), \qquad \Pi(\exp tX) = e^{t\,d\Pi(X)}.$$

If Π is unitary, then $d\Pi(X)^* = -d\Pi(X)$. Every vector can be differentiated.

For compact G, the Peter–Weyl theorem (Lecture 10) decomposes each unitary representation into finite-dimensional subrepresentations. That theory rests on averaging. The Haar integral of a compact group has finite total mass, so Lecture 5 can average an inner product, and Lecture 10 can average an operator. A noncompact Lie group also has an invariant integral, but its total mass is infinite (Lecture 16). So it has no normalised finite Haar integral, and these averaging arguments are no longer available in general.

The theory changes in the following ways. First, a finite-dimensional representation need not preserve any inner product. For example, for $m \geq 1$ and $t \neq 0$, the matrix $\operatorname{diag}(e^t, e^{-t}) \in \mathrm{SL}(2,\mathbb{R})$ acts on the module V_m with the eigenvalue e^{mt}, and a unitary operator has eigenvalues of modulus one. Lecture 23 proves more: every finite-dimensional unitary representation of $\mathrm{SL}(2,\mathbb{R})$ is trivial. So a nontrivial unitary representation of this group is infinite-dimensional.

This is not a property of every noncompact group. The group $\mathbb{R}$ has the one-dimensional unitary characters $x \mapsto e^{icx}$. It shows another change. Its translation representation on $L^2(\mathbb{R})$ is unitary, and it has no nonzero finite-dimensional invariant subspace (Proposition 15.4 below). So a unitary representation need not contain any finite-dimensional piece.

Finally, on an infinite-dimensional space the derivative of a representation may be defined only on a proper subspace. We begin with this point, on translations. Lecture 16 and Lecture 17 then build Haar measure and induction without compactness. We apply these tools to the Heisenberg group, to $\mathrm{SL}(2,\mathbb{R})$, and to the Lorentz and Poincaré groups. Lecture 26 and Lecture 27 return to compact groups, and Lecture 28 returns to Lecture 14.

The basic example is the translation representation on $L^2(\mathbb{R})$, so we first construct that space.

Let $C_c(\mathbb{R})$ be the space of continuous functions $\mathbb{R} \to \mathbb{C}$ with compact support, with inner product

$$\langle f, g \rangle = \int_{\mathbb{R}} \overline{f(x)}\, g(x)\, dx.$$

Its Hilbert completion is $L^2(\mathbb{R})$.

We want the indicator $1_{[0,1]}$ as a vector of $L^2(\mathbb{R})$, because it will be our example of a vector with no derivative. As in Lecture 10, a general element of a completion has no pointwise values. So we say which functions define vectors. A *jump* of a function $f\colon \mathbb{R} \to \mathbb{C}$ is a point where f is discontinuous but has both one-sided limits. We call f *piecewise continuous* if it is continuous except at finitely many jumps in each bounded interval. If moreover the improper integral $\int_{\mathbb{R}} |f|^2\, dx$ is finite, we call f *square integrable*. Such a function f defines a vector of $L^2(\mathbb{R})$, and for two such functions f and $\tilde{f}$,

$$\|f\|^2 = \int_{\mathbb{R}} |f|^2\, dx, \qquad \langle f, \tilde{f} \rangle = \int_{\mathbb{R}} \bar{f}\tilde{f}\, dx.$$

Changing f at finitely many points does not change the vector. § 15.6 constructs this vector and proves these formulas.

For $t \in \mathbb{R}$ and $f \in C_c(\mathbb{R})$ put

$$(\lambda(t)f)(x) = f(x - t).$$

Then $\lambda(s)\lambda(t) = \lambda(s+t)$, and $\|\lambda(t)f\| = \|f\|$ by substitution. So each $\lambda(t)$ extends to a unitary operator on $L^2(\mathbb{R})$ with inverse $\lambda(-t)$. On the piecewise continuous functions above, $\lambda(t)$ is still translation.

We identify the Lie algebra of $\mathbb{R}$ with $\mathbb{R}$. With this identification the exponential map $\mathbb{R} \to \mathbb{R}$ is the identity, so $d\lambda(1)f$ is the derivative of $t \mapsto \lambda(t)f$ at $t = 0$. Formally $\frac{d}{dt}\lambda(t)f\big|_{t=0} = -f'$. This formula has no meaning for a general vector. In the domain example, Example 15.16, the difference quotients of $1_{[0,1]}$ are unbounded. So $d\lambda(1)$ needs a stated domain.

15.2 Strong continuity

Definition 15.1. Let $\mathcal{H}$ be a complex Hilbert space and $\mathrm{U}(\mathcal{H})$ its group of unitary operators. A *unitary representation* of a Lie group G on $\mathcal{H}$ is a homomorphism $\Pi\colon G \to \mathrm{U}(\mathcal{H})$ such that the map $G \to \mathcal{H}$, $g \mapsto \Pi(g)v$, is continuous for every $v \in \mathcal{H}$. This condition is *strong continuity*. The homomorphism is *norm continuous* if $g \mapsto \Pi(g)$ is continuous for the operator norm.

Lemma 15.2. *Let $\Pi\colon G \to \mathrm{U}(\mathcal{H})$ be a homomorphism. Let $D \subset \mathcal{H}$ be dense, and suppose $g \mapsto \Pi(g)w$ is continuous at e for every $w \in D$. Then Π is strongly continuous.*

Proof. Let $v \in \mathcal{H}$ and $w \in D$. Since $\Pi(g)$ is unitary,

$$\|\Pi(g)v - v\| \le \|\Pi(g)(v-w)\| + \|\Pi(g)w - w\| + \|w - v\| = 2\|v-w\| + \|\Pi(g)w - w\|.$$

Given $\varepsilon > 0$, choose w with $\|v - w\| < \varepsilon$. Then $\limsup_{g\to e} \|\Pi(g)v - v\| \le 2\varepsilon$. So $g \mapsto \Pi(g)v$ is continuous at e. At another point h we use

$$\|\Pi(g)v - \Pi(h)v\| = \|\Pi(h^{-1}g)v - v\|,$$

and $h^{-1}g \to e$ as $g \to h$. ∎

Proposition 15.3. *The translations λ form a unitary representation of $\mathbb{R}$ on $L^2(\mathbb{R})$.*

Proof. Only strong continuity remains. We apply the dense-set criterion, Lemma 15.2, with $D = C_c(\mathbb{R})$. Let $f \in C_c(\mathbb{R})$ vanish outside $[-R, R]$. For $|t| \le 1$ the function $\lambda(t)f - f$ vanishes outside $[-R-1, R+1]$. Hence

$$\|\lambda(t)f - f\|^2 \le (2R+2) \sup_{x\in\mathbb{R}} |f(x-t) - f(x)|^2.$$

Since f is continuous with compact support, it is uniformly continuous. Hence the right side tends to 0 as $t \to 0$, and the dense-set criterion gives strong continuity. ∎

By the Peter–Weyl theorem (Lecture 10), every unitary representation of a compact group splits into finite-dimensional pieces. The next proposition shows that the unitary representation λ of the noncompact group $\mathbb{R}$ contains no such piece at all.

Proposition 15.4. *The representation λ has no nonzero finite-dimensional invariant subspace.*

Proof. First we show that matrix coefficients decay. Let $f, g \in C_c(\mathbb{R})$. For large $|t|$ the supports of g and $\lambda(t)f$ are disjoint, so $\langle g, \lambda(t)f\rangle = 0$. Now let $f, g \in L^2(\mathbb{R})$, and choose $f_0, g_0 \in C_c(\mathbb{R})$ close to f and g in norm. Since $\lambda(t)$ is unitary,

$$|\langle g, \lambda(t)f\rangle - \langle g_0, \lambda(t)f_0\rangle| \le \|g - g_0\|\,\|f\| + \|g_0\|\,\|f - f_0\|.$$

Hence $\langle g, \lambda(t)f\rangle \to 0$ as $|t| \to \infty$, for all $f, g \in L^2(\mathbb{R})$.

Suppose that $W \neq 0$ is a finite-dimensional invariant subspace, with orthonormal basis $e_1, \dots, e_k$. Let $f \in W$ be a unit vector. Then $\lambda(t)f$ is a unit vector of W, so

$$\sum_{j=1}^{k} |\langle e_j, \lambda(t)f \rangle|^2 = 1 \qquad \text{for every } t \in \mathbb{R}.$$

Each of the finitely many terms tends to 0 as $|t| \to \infty$. This is a contradiction. ∎

Norm continuity would be too strong a requirement, since translations are not norm continuous. Indeed, Exercise 15.1 shows that $\|\lambda(t) - I\| = 2$ for every $t \neq 0$. Strong continuity requires continuity separately for each vector. It does not require this continuity to be uniform over unit vectors.

Figure 15.1 shows the difference between a fixed vector and a vector chosen after t. For the fixed step $1_{[0,1]}$ and small $t > 0$, the translate differs from the step only on two slivers of total length $2t$. So $\|\lambda(t)1_{[0,1]} - 1_{[0,1]}\| \to 0$ as $t \to 0$. The comb of Exercise 15.1 depends on t, since its teeth have width t. Translation by t reverses its sign on the overlap, which keeps the ratio $\|\lambda(t)f - f\|/\|f\|$ away from 0 for every t. This does not contradict strong continuity, because strong continuity fixes the vector first.

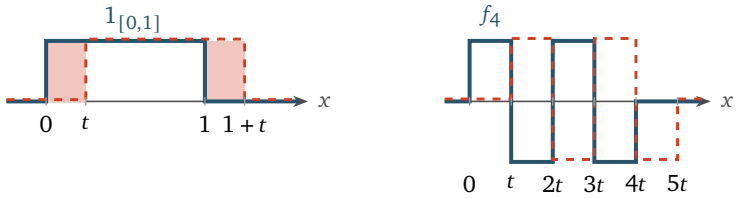


Figure 15.1: Two vectors, heavy, and their translates by t, dashed. The first panel shows the fixed step $1_{[0,1]}$, with the two slivers shaded. The second panel shows the comb f_4 of Exercise 15.1, whose teeth have width t.

15.3 Smooth vectors and the derived representation

A map from an open subset of $\mathbb{R}^n$ to $\mathcal{H}$ is *smooth* if all its partial derivatives of all orders exist as limits in the norm of $\mathcal{H}$ and are continuous. A map $G \to \mathcal{H}$ is smooth if it is smooth in every chart.

Definition 15.5. Let Π be a unitary representation of G on $\mathcal{H}$. A vector $v \in \mathcal{H}$ is *smooth* if $g \mapsto \Pi(g)v$ is a smooth map $G \to \mathcal{H}$. We write $\mathcal{H}^\infty$ for the set of smooth vectors. For $v \in \mathcal{H}^\infty$ and $X \in \mathfrak{g}$ put

$$d\Pi(X)v = \frac{d}{dt}\Big|_{t=0} \Pi(\exp tX)v = \lim_{t \to 0} \frac{\Pi(\exp tX)v - v}{t}.$$

The set $\mathcal{H}^\infty$ is a subspace of $\mathcal{H}$, since linear combinations of smooth maps $G \to \mathcal{H}$ are smooth. The limit defining $d\Pi(X)v$ exists because $t \mapsto \Pi(\exp tX)v$ is the smooth curve $t \mapsto \exp tX$ followed by the smooth map in the definition of a smooth vector.

Proposition 15.6. *Let Π be a unitary representation of a Lie group G on $\mathcal{H}$, and let $g \in G$, $X \in \mathfrak{g}$.*

(a) $\Pi(g)\mathcal{H}^\infty \subset \mathcal{H}^\infty$ *and* $d\Pi(X)\mathcal{H}^\infty \subset \mathcal{H}^\infty$.
(b) $\mathcal{H}^\infty$ *is dense in* $\mathcal{H}$.
(c) $X \mapsto d\Pi(X)$ *is a representation of* $\mathfrak{g}$ *on* $\mathcal{H}^\infty$, *and* $d\Pi(\mathrm{Ad}(g)X) = \Pi(g)\, d\Pi(X)\, \Pi(g)^{-1}$ *on* $\mathcal{H}^\infty$.
(d) *For* $v, w \in \mathcal{H}^\infty$ *we have* $\langle d\Pi(X)v, w\rangle = -\langle v, d\Pi(X)w\rangle$.

Proof. We cite Knapp, *Representation Theory of Semisimple Groups*, Sections III.3–4, for the invariance under $d\Pi(X)$ in (a), the density in (b), and the representation property and Ad formula in (c). We prove the rest.

For the invariance under $\Pi(g)$ in (a), let $v \in \mathcal{H}^\infty$, $h \in G$ and $F(g) = \Pi(g)v$. Then $\Pi(g)\Pi(h)v = F(gh)$. Right translation by h is smooth on G, so $g \mapsto F(gh)$ is smooth.

For the skew-symmetry in (d), unitarity gives

$$\langle \Pi(\exp tX)v, w\rangle = \langle v, \Pi(\exp(-tX))w\rangle.$$

Subtract $\langle v, w\rangle$ from both sides and divide by $t \neq 0$:

$$\left\langle \frac{\Pi(\exp tX)v - v}{t},\, w \right\rangle = -\left\langle v,\, \frac{\Pi(\exp(-tX))w - w}{-t} \right\rangle.$$

The inner product is continuous, so letting $t \to 0$ gives (d). ∎

The omitted density step in (b) is the hard one, so we sketch it. The sketch uses a vector-valued integral against a left Haar measure dg on G. For compact G, Haar measure is introduced in Lecture 5. For noncompact G it is introduced in Lecture 16, so until then we cite Knapp for this integral. For $\varphi \in C_c^\infty(G)$ and $v \in \mathcal{H}$ form the *Gårding vector*

$$\Pi(\varphi)v = \int_G \varphi(g)\, \Pi(g)v\, dg.$$

Let $h \in G$. Since $\Pi(h)$ is bounded, it passes inside the integral. Left invariance of dg then allows the substitution $g \mapsto h^{-1}g$. It gives

$$\Pi(h)\Pi(\varphi)v = \int_G \varphi(g)\, \Pi(hg)v\, dg = \int_G \varphi(h^{-1}g)\, \Pi(g)v\, dg.$$

Now h appears only inside the smooth scalar function $(h, g) \mapsto \varphi(h^{-1}g)$. Differentiating under the integral in local coordinates of h shows that $h \mapsto \Pi(h)\Pi(\varphi)v$ is smooth on G. So $\Pi(\varphi)v$ is a smooth vector. If $\varphi_n \geq 0$, $\int_G \varphi_n\, dg = 1$ and the supports shrink to e, then

$$\|\Pi(\varphi_n)v - v\| \leq \sup_{g \in \operatorname{supp}\varphi_n} \|\Pi(g)v - v\| \longrightarrow 0$$

by strong continuity. Exercise 15.4 carries out the argument for $G = \mathbb{R}$ with Riemann integrals.

15.4 Unbounded operators and Stone's theorem

On $L^2(\mathbb{R})$ the operator $-d/dx$ is defined only on a subspace. We need language for such operators.

Definition 15.7. An *operator* A on $\mathcal{H}$ is a linear map $A\colon D(A) \to \mathcal{H}$ defined on a subspace $D(A) \subset \mathcal{H}$, its *domain*. It is *densely defined* if $D(A)$ is dense. We write $B \subset A$ if $D(B) \subset D(A)$ and $Bv = Av$ for $v \in D(B)$.

Definition 15.8. Let A be densely defined. The domain of the *adjoint* A^* is the set of $w \in \mathcal{H}$ for which some $u \in \mathcal{H}$ satisfies

$$\langle w, Av\rangle = \langle u, v\rangle \qquad \text{for all } v \in D(A),$$

and then $A^*w = u$. The operator A is *skew-symmetric* if $\langle Av, w\rangle = -\langle v, Aw\rangle$ for all $v, w \in D(A)$. It is *skew-adjoint* if $A^* = -A$, including $D(A^*) = D(A)$, and *self-adjoint* if $A^* = A$, including $D(A^*) = D(A)$.

Since $D(A)$ is dense, the vector u is unique. Skew-symmetry of A means exactly that $-A \subset A^*$.

Suppose $B \subset A$ and B is densely defined. Then $D(A) \supset D(B)$ is dense too, so both adjoints exist. In this case $A^* \subset B^*$. Indeed, the defining identity for A^*w holds in particular for $v \in D(B)$.

Lemma 15.9. *Let B be densely defined and skew-symmetric. If $v_n \in D(B)$, $v_n \to 0$ and $Bv_n \to u$, then $u = 0$.*

Proof. For every $w \in D(B)$, continuity of the inner product and skew-symmetry of B give

$$\langle w, u\rangle = \lim_n \langle w, Bv_n\rangle = -\lim_n \langle Bw, v_n\rangle = 0.$$

So u is orthogonal to the dense subspace $D(B)$, and $u = 0$. ∎

Definition 15.10. Let B be densely defined and skew-symmetric. The domain of its *closure* $\bar{B}$ is the set of $v \in \mathcal{H}$ for which there are $v_n \in D(B)$ with $v_n \to v$ and (Bv_n) convergent. Then $\bar{B}v = \lim_n Bv_n$.

The value $\bar{B}v$ does not depend on the sequence. Indeed, the difference of two such sequences satisfies the hypotheses of the limit lemma for skew-symmetric operators, Lemma 15.9.

Definition 15.11. Let $U\colon \mathbb{R} \to \mathrm{U}(\mathcal{H})$ be a unitary representation. Its *generator* A has domain

$$D(A) = \left\{ v \in \mathcal{H} : \lim_{t \to 0} \frac{U(t)v - v}{t} \text{ exists in } \mathcal{H} \right\},$$

and Av is this limit.

Theorem 15.12 (Stone). *Let $U\colon \mathbb{R} \to \mathrm{U}(\mathcal{H})$ be a strongly continuous unitary representation with generator A.*

(a) A is densely defined and skew-adjoint. Moreover $U(t)D(A) \subset D(A)$ and $AU(t)v = U(t)Av$ for $v \in D(A)$.

(b) *Let $D \subset D(A)$ be a dense subspace with $U(t)D \subset D$ for all t. Then $A|_D$ is skew-symmetric, and its closure is A.*

(c) *Every skew-adjoint operator is the generator of exactly one strongly continuous unitary representation of $\mathbb{R}$, written $t \mapsto e^{tA}$.*

We cite most of this theorem and prove only the invariance statement in (a). The sources are Hall, *Quantum Theory for Mathematicians*, Section 10.2, and Reed–Simon, *Methods of Modern Mathematical Physics I*, Section VIII.4. The section on sources and hypotheses at the end of this lecture matches their conventions with ours. For $v \in D(A)$, since $U(t)$ is bounded and commutes with $U(h)$,

$$\frac{U(h)U(t)v - U(t)v}{h} = U(t)\,\frac{U(h)v - v}{h} \longrightarrow U(t)Av \qquad (h \to 0).$$

So $U(t)v \in D(A)$ and $AU(t)v = U(t)Av$.

Our physical convention writes $U(t) = e^{-itH}$ with a self-adjoint operator H, so $H = iA$.

Corollary 15.13. *Let Π be a unitary representation of a Lie group G on $\mathcal{H}$. Let $X \in \mathfrak{g}$, and let B_X be the generator of $t \mapsto \Pi(\exp tX)$. Then $\mathcal{H}^\infty \subset D(B_X)$ and $B_X = d\Pi(X)$ on $\mathcal{H}^\infty$. The closure of $d\Pi(X)$ with domain $\mathcal{H}^\infty$ is the skew-adjoint operator B_X, and $\Pi(\exp tX) = e^{t\,\overline{d\Pi(X)}}$.*

Proof. The map $t \mapsto \exp tX$ is a continuous homomorphism $\mathbb{R} \to G$. Composing it with Π gives a homomorphism $\mathbb{R} \to \mathrm{U}(\mathcal{H})$. For each v, the map $t \mapsto \Pi(\exp tX)v$ is a continuous map followed by the continuous map $g \mapsto \Pi(g)v$. So $t \mapsto \Pi(\exp tX)$ is a strongly continuous unitary representation of $\mathbb{R}$, and its generator B_X is defined.

The first two claims restate the definition of $d\Pi(X)$. By the invariance and density statements of Proposition 15.6, $\mathcal{H}^\infty$ is dense and invariant under every $\Pi(\exp tX)$. The dense invariant-domain criterion of Theorem 15.12(b), applied with $D = \mathcal{H}^\infty$, shows that the closure of $d\Pi(X)$ is B_X. The uniqueness statement of Theorem 15.12(c) then gives $\Pi(\exp tX) = e^{t\,\overline{d\Pi(X)}}$. ∎

The generator B_X can have a larger domain than $\mathcal{H}^\infty$. That domain can depend on X, and B_X need not preserve it. For example, Exercise 15.2 gives a vector g in the domain of the generator of translations whose image is outside that domain. Brackets $[d\Pi(X), d\Pi(Y)]$ need one subspace on which every $d\Pi(X)$ is defined and which every $d\Pi(X)$ preserves. By Proposition 15.6, the space $\mathcal{H}^\infty$ has both properties.

In the physical convention above, the *physical generator* of $t \mapsto \Pi(\exp tX)$ is $A_X = iB_X$, so $\Pi(\exp tX) = e^{-itA_X}$. Since B_X is skew-adjoint, A_X is self-adjoint. For translations and $X = 1 \in \mathbb{R}$, the generator B_1 is the operator that the next section calls A. The subscript matters: the physical generator $A_1 = iA$ is self-adjoint, while A is skew-adjoint. With $\hbar = 1$, the operator A_1 is the momentum operator P, formally $-i\,d/dx$.

15.5 Translations on $L^2(\mathbb{R})$

The *Schwartz space* $\mathcal{S}(\mathbb{R})$ is the space of smooth $f\colon \mathbb{R} \to \mathbb{C}$ such that

$$\sup_{x \in \mathbb{R}} |x^k f^{(j)}(x)| < \infty \qquad \text{for all } j, k \geq 0.$$

It contains $C_c^\infty(\mathbb{R})$ and $e^{-x^2/2}$. Let A be the generator of λ.

Lemma 15.14. *Let $f \in C^2(\mathbb{R})$ with f and f' square integrable. Suppose $|f''(y)| \le C(1+y^2)^{-1}$ for all y. Then $f \in D(A)$ and $Af = -f'$.*

Proof. Fix $0 < |h| \le 1$. Taylor's formula with integral remainder, for $s \mapsto f(x - sh)$ on $[0,1]$, gives

$$f(x-h) = f(x) - hf'(x) + h^2 \int_0^1 (1-s) f''(x-sh)\, ds.$$

So the error function is

$$q_h(x) = \frac{f(x-h) - f(x)}{h} + f'(x) = h \int_0^1 (1-s) f''(x-sh)\, ds,$$

and $|q_h(x)| \le \frac{|h|}{2} \sup_{|y-x|\le 1} |f''(y)|$. If $|y - x| \le 1$, then $x^2 \le (|y|+1)^2 \le 2y^2 + 2$, so $1 + x^2 \le 3(1+y^2)$. Therefore

$$|q_h(x)| \le \frac{3C|h|}{2(1+x^2)}, \qquad \|q_h\|^2 \le \frac{9C^2h^2}{4} \int_{\mathbb{R}} \frac{dx}{(1+x^2)^2} = \frac{9\pi C^2 h^2}{8}.$$

Thus $\left\| \frac{\lambda(h)f - f}{h} - (-f') \right\| = \|q_h\| \to 0$. ∎

The essential hypothesis is the weighted bound $|f''(y)| \le C(1+y^2)^{-1}$. The difference quotient in the proof converges pointwise for every C^1 function f. Pointwise convergence alone does not give convergence in norm. The weighted bound makes the error q_h at most $|h|$ times a fixed square-integrable function, so the norm of q_h is $O(|h|)$.

Proposition 15.15. *The space $\mathcal{S}(\mathbb{R})$ is dense in $L^2(\mathbb{R})$, invariant under every $\lambda(t)$, and contained in $L^2(\mathbb{R})^\infty$. For $f \in \mathcal{S}(\mathbb{R})$ we have $d\lambda(1)f = -f'$. The closure of $B = -d/dx$ with domain $\mathcal{S}(\mathbb{R})$ is the skew-adjoint generator A.*

Proof. Invariance. Put $y = x - t$. Expanding $x^k = (y+t)^k$ bounds $|x^k f^{(j)}(x-t)|$ by a finite sum of terms $\binom{k}{i}|t|^{k-i}|y^i f^{(j)}(y)|$.

Smoothness. Each $f \in \mathcal{S}(\mathbb{R})$ satisfies the hypotheses of the Taylor-remainder lemma, Lemma 15.14. Indeed $(1+y^2)f$, $(1+y^2)f'$ and $(1+y^2)f''$ are bounded. So f and f' are square integrable, and $|f''| \le C(1+y^2)^{-1}$. In particular $\mathcal{S}(\mathbb{R}) \subset L^2(\mathbb{R})$. By the lemma, $t \mapsto \lambda(t)f$ is differentiable at 0 with derivative $-f'$. At any t_0,

$$\frac{\lambda(t_0+h)f - \lambda(t_0)f}{h} = \lambda(t_0)\, \frac{\lambda(h)f - f}{h} \longrightarrow \lambda(t_0)(-f'),$$

because $\lambda(t_0)$ is bounded. The derivative $t \mapsto \lambda(t)(-f')$ is continuous by the strong continuity of translations, Proposition 15.3. Since $-f' \in \mathcal{S}(\mathbb{R})$, we may repeat. The k-th derivative is $t \mapsto \lambda(t)\big((-1)^k f^{(k)}\big)$, so f is a smooth vector.

Density. Fix $\varphi \in C_c^\infty(\mathbb{R})$ with $\varphi \ge 0$, support in $[-1,1]$ and $\int \varphi = 1$. Put $\varphi_n(t) = n\varphi(nt)$. For $f \in C_c(\mathbb{R})$ vanishing outside $[-R, R]$, let

$$f_n(x) = \int_{\mathbb{R}} \varphi_n(x-y) f(y)\, dy = \int_{\mathbb{R}} \varphi_n(t) f(x-t)\, dt.$$

The first form shows $f_n \in C_c^\infty(\mathbb{R}) \subset \mathcal{S}(\mathbb{R})$, with support in $[-R-1, R+1]$. The second form gives

$$|f_n(x) - f(x)| \le \sup_{|t|\le 1/n} |f(x-t) - f(x)|,$$

which tends to 0 uniformly in x because f is uniformly continuous. Since f_n and f vanish outside $[-R-1, R+1]$, it follows that $f_n \to f$ in $L^2(\mathbb{R})$. Since $C_c(\mathbb{R})$ is dense in $L^2(\mathbb{R})$, so is $\mathcal{S}(\mathbb{R})$.

Closure. By the three steps above, $D = \mathcal{S}(\mathbb{R})$ is a dense invariant subspace of $D(A)$. So the dense invariant-domain criterion of Theorem 15.12(b) shows that the closure of B is A. ∎

The function f_n is the Gårding vector $\lambda(\varphi_n)f$ written pointwise. By analogy with the compact groups of Lecture 11, it is the convolution $\varphi_n * f$.

Example 15.16. The vectors w and h, and the operator $B = -d/dx$ on $\mathcal{S}(\mathbb{R})$, show where the domains lie.

(i) Let $w(x) = (1+x^2)^{-1}$. Then

$$w'(x) = \frac{-2x}{(1+x^2)^2}, \qquad w''(x) = \frac{6x^2-2}{(1+x^2)^3}.$$

Since $|6x^2 - 2| \le 6(1+x^2)$, we get $|w''(x)| \le 6(1+x^2)^{-2} \le 6(1+x^2)^{-1}$. Both w and w' are square integrable. By the Taylor-remainder lemma, Lemma 15.14, $w \in D(A)$ and

$$(Aw)(x) = \frac{2x}{(1+x^2)^2}.$$

But $x^2 w(x) \to 1$, so $x^4 w(x)$ is unbounded and $w \notin \mathcal{S}(\mathbb{R})$.

(ii) The operator $B = -d/dx$ on $\mathcal{S}(\mathbb{R})$ is the restriction of $d\lambda(1)$ to $\mathcal{S}(\mathbb{R})$ by Proposition 15.15. So B is skew-symmetric by the skew-symmetry of $d\lambda(1)$ on smooth vectors, Proposition 15.6(d). It is not skew-adjoint. Indeed $B \subset A$, the domain $\mathcal{S}(\mathbb{R})$ is dense, and $A^* = -A$. So $-A = A^* \subset B^*$. Hence $w \in D(A) \subset D(B^*)$, while $w \notin D(B)$. So $D(B^*) \neq D(B)$, and therefore $B^* \neq -B$.

(iii) Let $h = 1_{[0,1]}$. For $0 < t \le 1$, up to finitely many points,

$$\lambda(t)h - h = 1_{[t,1+t]} - 1_{[0,1]} = 1_{(1,1+t]} - 1_{[0,t)}, \qquad \|\lambda(t)h - h\|^2 = 2t.$$

For $-1 \le t < 0$ the same computation gives $2|t|$. Therefore

$$\left\| \frac{\lambda(t)h - h}{t} \right\| = \sqrt{\frac{2}{|t|}} \longrightarrow \infty.$$

Convergent quotients would stay bounded near 0. So $h \notin D(A)$. Smooth vectors lie in $D(A)$ by Corollary 15.13, so also $h \notin L^2(\mathbb{R})^\infty$.

Together these give

$$\mathcal{S}(\mathbb{R}) \subset L^2(\mathbb{R})^\infty \subset D(A), \qquad w \in D(A) \setminus \mathcal{S}(\mathbb{R}), \qquad h \in L^2(\mathbb{R}) \setminus D(A).$$

Exercise 15.2 gives a tent function in $D(A)$ that is not a smooth vector. All derivatives of w satisfy the hypotheses of the Taylor-remainder lemma, Lemma 15.14, so w is a smooth vector that is not in $\mathcal{S}(\mathbb{R})$. So both inclusions are strict. The momentum operator $P = iA$ has domain $D(A)$, not $\mathcal{S}(\mathbb{R})$.

Remark 15.17. We cite Plancherel's theorem (Folland, *Real Analysis*, second edition, Chapter 8). The Fourier transform $\hat{f}(p) = (2\pi)^{-1/2} \int_{\mathbb{R}} f(x) e^{-ipx}\, dx$ extends to a unitary operator of $L^2(\mathbb{R})$, and it turns $\lambda(t)$ into multiplication by e^{-ipt}. As in Exercise 15.3, $D(A)$ is the set of f with $\int_{\mathbb{R}} p^2 |\hat{f}(p)|^2\, dp < \infty$, and P acts as multiplication by p. In quantum mechanics, $D(A)$ is the set of states for which P^2 has a finite expectation value.

This describes the three vectors above. The step $h = 1_{[0,1]}$ has $|\hat{h}(p)|^2 = 2\sin^2(p/2)/(\pi p^2)$, so $\int p^2 |\hat{h}|^2\, dp$ diverges. The tent g of Exercise 15.2 has $|\hat{g}(p)|^2 = 8\sin^4(p/2)/(\pi p^4)$, so $\int p^2 |\hat{g}|^2\, dp$ converges, while $\int p^4 |\hat{g}|^2\, dp$ diverges. The function w has $\hat{w}(p) = \sqrt{\pi/2}\, e^{-|p|}$, so all these integrals converge. Finally, $\|\lambda(t) - I\| = \sup_p |e^{-ipt} - 1| = 2$ for $t \neq 0$, which agrees with Exercise 15.1.

15.6 Piecewise continuous functions as vectors

We now construct the vector of a square integrable function, as stated in § 15.1. We first treat functions that vanish outside a bounded interval. Then we pass to square integrable functions by truncation.

Let f be piecewise continuous and vanish outside a bounded interval. Then f has finitely many jumps $x_1 < \cdots < x_J$, and the one-sided limits at these jumps make f bounded, say $|f| \le M$. Choose $\delta > 0$ so small that the intervals $[x_j - \delta, x_j + \delta]$ are disjoint. Let g_δ equal f outside these intervals, and let g_δ be linear on each of them, with the values of f at the two endpoints. Then $g_\delta \in C_c(\mathbb{R})$ and $|g_\delta| \le M$. So $|f - g_\delta| \le 2M$, and $f - g_\delta$ vanishes outside J intervals of length 2δ. Hence

$$\int_{\mathbb{R}} |f - g_\delta|^2\, dx \le 8JM^2\delta.$$

For piecewise continuous functions u and $\tilde{u}$ that vanish outside a bounded interval, write $\|u\|_2 = \big(\int_{\mathbb{R}} |u|^2\, dx\big)^{1/2}$. On $C_c(\mathbb{R})$ this is the norm of $L^2(\mathbb{R})$. The Cauchy–Schwarz inequality for Riemann integrals, $\int_{\mathbb{R}} |\bar{u}\tilde{u}|\, dx \le \|u\|_2 \|\tilde{u}\|_2$, gives the triangle inequality for $\|\cdot\|_2$.

Take $\delta_n \to 0$ and $g_n = g_{\delta_n}$, so that $\|f - g_n\|_2 \to 0$. Since $\|g_n - g_m\|_2 \le \|g_n - f\|_2 + \|f - g_m\|_2$, the sequence (g_n) is Cauchy in $L^2(\mathbb{R})$. We call its limit the vector f. By the same inequality, every sequence in $C_c(\mathbb{R})$ that approximates f in $\|\cdot\|_2$ has this limit. Sums and scalar multiples of approximating sequences approximate sums and scalar multiples, so the vector depends linearly on f. By the triangle inequality, the vector f has norm $\|f\| = \lim_n \|g_n\|_2 = \|f\|_2$.

Let $\tilde{f}$ be a second such function, approximated by $\tilde{g}_n \in C_c(\mathbb{R})$. By continuity of the inner product, $\langle f, \tilde{f}\rangle = \lim_n \int_{\mathbb{R}} \overline{g_n}\, \tilde{g}_n\, dx$. Also

$$\int_{\mathbb{R}} \overline{g_n}\, \tilde{g}_n\, dx - \int_{\mathbb{R}} \bar{f}\tilde{f}\, dx = \int_{\mathbb{R}} \overline{(g_n - f)}\, \tilde{g}_n\, dx + \int_{\mathbb{R}} \bar{f}\, (\tilde{g}_n - \tilde{f})\, dx.$$

By the Cauchy–Schwarz inequality, the right side has absolute value at most $\|g_n - f\|_2 \|\tilde{g}_n\|_2 + \|f\|_2 \|\tilde{g}_n - \tilde{f}\|_2$. This tends to 0, because $\|\tilde{g}_n\|_2$ stays bounded. So $\langle f, \tilde{f}\rangle = \int_{\mathbb{R}} \bar{f}\tilde{f}\, dx$.

Now let f be square integrable. Each truncation $f \cdot 1_{[-n,n]}$ is piecewise continuous and vanishes outside $[-n, n]$, so it defines a vector. The difference of two truncations is $f \cdot$

$1_{\{n<|x|\le m\}}$. By linearity and the norm formula, the truncations form a Cauchy sequence, because

$$\left\| f\cdot 1_{[-m,m]} - f\cdot 1_{[-n,n]}\right\|^2 = \int_{n<|x|\le m} |f|^2\,dx \longrightarrow 0 \qquad (m>n\to\infty).$$

Their limit is the vector f. For square integrable f and $\tilde f$, letting $n\to\infty$ in the formulas for the truncations gives the two formulas of § 15.1. The second improper integral converges absolutely, since $\int_{-n}^{n}|\bar f\tilde f|\,dx \le \|f\|\,\|\tilde f\|$ by the Cauchy–Schwarz inequality.

Changing f at finitely many points does not change the vector. Indeed, the difference of the two functions is nonzero at only finitely many points, so its vector has norm 0.

15.7 Self-checks

1. Let $\dim\mathcal{H}<\infty$. What are $\mathcal{H}^\infty$ and $D(B_X)$?
   Answer. Both equal $\mathcal{H}$. The homomorphism $\Pi\colon G\to \mathrm{U}(\mathcal{H})$ is continuous, hence smooth (Lecture 3), so every vector is smooth. Since $d\Pi(X)$ is then defined on all of $\mathcal{H}$, its closure B_X also has domain $\mathcal{H}$.
2. Let $f(x)=xe^{-x^2/2}$. Compute Af and $\langle f, Af\rangle$.
   Answer. Since $f\in\mathcal{S}(\mathbb{R})$, the Schwartz-space proposition, Proposition 15.15, gives $Af=-f'=(x^2-1)e^{-x^2/2}$. Then $\langle f,Af\rangle=\int_{\mathbb{R}} x(x^2-1)e^{-x^2}\,dx=0$, because the integrand is odd. This is consistent with skew-symmetry, which forces $\langle f,Af\rangle$ to be purely imaginary.
3. Let D be the space of smooth functions with compact support in $(0,\infty)$. Then $D\subset D(A)$. Does the dense invariant-domain criterion of Theorem 15.12(b) apply to D?
   Answer. No. The space D is not dense, since every element is orthogonal to $1_{[-1,0]}\neq 0$. It is not invariant either. Given $t<0$, choose a nonzero $f\in D$ supported in $(0,|t|)$. Then $\lambda(t)f$ is supported in $(t,0)$, so $\lambda(t)f\notin D$.
4. Let $D=C_c^\infty(\mathbb{R}\setminus\{0\})$. This space is dense in $L^2(\mathbb{R})$ and contained in $D(A)$, but it is not invariant. Is the closure of $A|_D$ equal to A?
   Answer. No. Let g be a C^1 function with g and g' square integrable and $g(x)\to 0$ as $x\to-\infty$. Then $|g(0)|^2=\int_{-\infty}^{0}(|g|^2)'\,dx\le 2\|g\|\,\|g'\|$. Suppose that $v_n\in D$ with $v_n\to w$ and $Av_n\to Aw$, where $w=(1+x^2)^{-1}$. Take $g=w-v_n$. Since $v_n(0)=0$ and $w(0)=1$, we get $1\le 2\|w-v_n\|\,\|w'-v_n'\|\to 0$, which is false. So $w\in D(A)$ is not in the domain of the closure. The invariance hypothesis in the criterion cannot be dropped.

15.8 Summary

Unitary representations are required to be strongly continuous, not norm continuous. The translation representation on $L^2(\mathbb{R})$ has no nonzero finite-dimensional invariant subspace. Smooth vectors form a dense invariant subspace $\mathcal{H}^\infty$. On it, $d\Pi$ is a representation of $\mathfrak{g}$ by skew-symmetric operators. Stone's theorem attaches a skew-adjoint generator to each strongly continuous unitary representation of $\mathbb{R}$. The closure of $d\Pi(X)$ on $\mathcal{H}^\infty$ is this generator. For translations, $-d/dx$ on $\mathcal{S}(\mathbb{R})$ is skew-symmetric but not skew-adjoint. Its closure has the strictly larger domain $D(A)$, and $D(A)$ is still not all of $L^2(\mathbb{R})$.

Sources and hypotheses

Hall, *Quantum Theory for Mathematicians*, Section 10.2, Theorem 10.15, gives Stone's theorem. Hall writes $U(t) = e^{itH_{\text{Hall}}}$ with a self-adjoint operator H_{Hall}, so $H_{\text{Hall}} = -iA$. In our physical convention $H = iA$, so $H_{\text{Hall}} = -H$. Reed–Simon, *Methods of Modern Mathematical Physics I*, Section VIII.4, Theorems VIII.10–11, gives the dense invariant-domain criterion in Theorem 15.12(b).

15.9 Reading

- Hall, *Quantum Theory for Mathematicians* (2013): unbounded self-adjoint operators and Stone's theorem, Section 10.2.
- Reed–Simon, *Methods of Modern Mathematical Physics I: Functional Analysis*, Section VIII.4, especially Theorems VIII.10–11 on dense invariant domains.
- Knapp, *Representation Theory of Semisimple Groups* (2001), Sections III.3–4: smooth vectors, Gårding vectors and the action of $\mathfrak{g}$.
- Etingof, MIT 18.757 lecture notes, Lectures 1–7: an alternative introduction to continuous representations of Lie groups.